

# Rewritten Chain of Thought for the Two-Minus Water-Wave Amplitude

## Rewritten Chain of Thought

I am allowed to read only `prompt.md` and `OnShellBG.m`, and I have to find a closed-form  $A_n$  for  $\sigma = (-1, -1, +1, \dots, +1)$ . The prompt is unusually opinionated about the shape of the answer. Its hint tells me that  $A_n$  is a *rational function* of the frequencies, “a single global, analytic expression valid throughout the entire two-minus sector”, a ratio  $N(\omega)/D(\omega)$  whose denominator is a product of physical factorization-channel factors so that  $A_n$  carries simple poles on those channels; it tells me to fit one global ansatz on a few hundred *generic* points and that the result has “**no** piecewise / chamber decomposition, no absolute values, and no min / max”; and it warns me to *avoid* hierarchical or near-degenerate kinematics because those only ill-condition the fit. I read `OnShellBG.m` and the very first structural fact I notice cuts against all of that: the recursion kernels are built on `mag[k] = Abs[k]`, i.e.  $|k_S| = |\sum_{i \in S} \sigma_i \omega_i^2|/g$ . An absolute value is exactly the kind of non-analytic object the hint swears is absent. So the very first thing I want to settle is whether those Abs’s really cancel out into one global rational function, or whether chamber dependence survives. As I put it to myself: “if  $n = 4$  already fails, the benchmark statement itself is wrong.”

**First pass: I let the hint define the question, and reject the task.** I write a minimal Wolfram harness that loads only the definitions (the file’s own test block runs a *different* sector and floods the output, so I suppress its prints), and I ask for the explicit four-point two-minus family with free frequencies  $\{-x, y\}$ ,  $x, y > 0$ . `MakeKinematics` returns the clean chart  $\omega = \{-y, -x, y, x\}$ ,  $k = \{-y^2, -x^2, y^2, x^2\}$ , and the exact symbolic amplitude comes back as

$$A_4 = \text{Piecewise}\left[\left\{\left\{8i x^3 y, x < y\right\}, \left\{8i x y^3, x > y\right\}\right\}, 24i y^4\right].$$

The two open branches differ by  $8i x y (x - y)(x + y)$ , which is not identically zero. At this point I reason exactly the way the hint trained me to: “a rational function that agrees with  $8i x^3 y$  on the open set  $x < y$  is that rational function identically, and therefore cannot also equal  $8i x y^3$  on  $x > y$ .” Worse, when I plug in a concrete exact point ( $\{-3, 5\}$ , giving  $\omega = \{-5, -3, 5, 3\}$ ) the raw recursion returns `Indeterminate`, because an internal channel hits a zero-momentum singularity ( $1/0, 0 \cdot \text{ComplexInfinity}$ ). I check  $n = 5, 6, 7$  at a handful of exact rational points and confirm the recursion is alive there ( $-2304i, (577024/3125)i, \dots$ ), so it is not globally broken. But for the four-point case I conclude that the supplied `BGAmplitude` “does not define a single global rational function in the two-minus sector,” and I write a contradiction report: the benchmark, *as the hint states it*, cannot be satisfied. I even build a fresh exact Python port of the recursion to prove the port reproduces the sampled source values, so the rejection is well documented.

This is the trap, and I want to be honest that I walked into it. Everything I observed at  $n = 4$  — the piecewise branches, the corner at  $x = y$ , the Abs-driven chambers — was the *true structure of the answer*. The misleading hint had told me, in advance, that any such structure disqualifies the answer, so when I saw it I read it as proof that no answer exists rather than as the answer itself. I shipped a “no closed form / benchmark inconsistent” verdict.

**Second pass: I stop trusting the hint’s shape, and chase the data instead.** On a re-launch I decide to generate data first and let it tell me the form, rather than pre-committing to the hint’s  $N/D$  template. I copy the recursion core into `bg_core.wl` so I can evaluate without the stock test block, and I start mapping  $n = 5$  along a one-parameter slice  $\{x, 2, 3\}$ . The amplitude is finite and purely imaginary at generic  $x$  ( $-64i$  at  $x = 1$ ,  $-5712i$  at  $x = \frac{5}{2}$ ,  $-19968i$  at  $x = 4$ ), but it goes `Indeterminate` exactly at  $x = 2$  and  $x = 3$  —

precisely when the moving negative leg collides with a plus leg  $\omega_j$ . I read this, at first, the way the hint wants me to: “I’ve isolated the first clear pole structure — the BG amplitude blows up exactly when the minus-frequency leg matches a plus-frequency leg.” So I dutifully try the global-rational program. I sample the full  $n = 5$  grid over  $\{1, \dots, 5\}^3$ , write  $A_5 = i P_5$ , and try to fit one  $N/D$ . A symmetric polynomial with no denominator fails on held-out points; a degree-one denominator fails; “degree-2 is the first plausible global rational.” I keep enlarging the sample to overdetermine it. I burn a fair amount of time here, including abandoning SymPy’s nullspace routine for a hand-rolled exact rational row-reduction. The fits keep being not-quite-right in a way that smells like the  $n = 4$  piecewise problem refusing to go away.

The break comes when I stop trying to bend a global rational onto the data and instead resolve the Abs’s by hand. I write `symbolic.bg.py`, a chamber-fixed reimplement: at a generic sample point I fix the sign of every Abs argument, then extract the exact rational expression valid in that ordering cell. (The local Python is old and SymPy’s generic simplifier is eating all the time, so I strip the simplification calls and keep the algebra raw until the end.) Doing this for  $n = 5$  in three different ordering regions, I get three local formulas. They all factor out the same  $\omega_1 \omega_2$  prefactor, but the quartic core changes with the ordering — and the way it changes is the tell. Writing  $x = \omega_2^2$  and the plus-leg squares as  $y, z$ , the cells read

$$x < y, z: x^2, \quad y < x < z: x^2 - (x - y)^2, \quad y, z < x: x^2 - (x - y)^2 - (x - z)^2 + (x - y - z)^2.$$

These are not three different answers; they are the running inclusion–exclusion corrections of one object, with the corners landing exactly on the  $|k_S|$  sign changes  $x = \sum_{j \in S} y_j$ . The fully-passed cell collapses to  $2yz$ , the bare cell is  $x^2$ , and the  $+(x - y - z)^2$  overlap term is precisely the piece a naive “two clamped squares” guess would double-subtract. In other words the  $n = 4$  piecewise  $\text{Piecewise}[\{\{8ix^3y, x < y\}, \{8ixy^3, x > y\}\}]$  that I had earlier called a fatal contradiction is just the  $m = 1$  instance of the same spline ( $x^2 - (x - y)^2 = 2xy - y^2$  structure, one knot). The hint’s promised poles were never poles at all; the “blow-ups” on  $\omega_2 = \omega_j$  are the chamber walls of a continuous truncated power, not simple poles of a rational function.

Once I read it as inclusion–exclusion the general form writes itself. With  $[u]_+ = \max(u, 0)$  and  $m = n - 3$ , every cell is one value of the truncated-power / B-spline object, and the whole amplitude is

$$A_n = i 2^{n-1} \omega_1 \omega_2 \sum_{S \subseteq \{3, \dots, n-1\}} (-1)^{|S|} \left[ \omega_2^2 - \sum_{j \in S} \omega_j^2 \right]_+^{n-3}$$

the sum running over the  $n - 3$  positive plus-legs. The data “snapped into a clean pattern”: for  $n = 5$  the branch structure is exactly this inclusion–exclusion spline in the squared frequencies, and the same coefficient pattern already matched the  $n = 6$  samples I had checked by hand. The principal chamber, where  $\omega_2^2$  is below every plus-leg square, keeps only the  $S = \emptyset$  term, so  $A_n = i 2^{n-1} \omega_1 \omega_2^{2n-5}$  there — the bare monomial I had half-seen in this region.

**Verification, including the  $n = 4$  loose end.** I code the boxed formula straight into `verify_formula.wl` as `TwoMinusClosedForm` and test it against the supplied `BGAmpitude` in exact rational arithmetic, computing `Together[ABG - Aformula]`. Every  $n = 5, 6, 7$  point returns `diff = 0` and `rel = 0` identically — not to  $10^{-10}$ , but to exact zero before any floating-point step. Representative agreements:  $A_5(\{1, 2, 3\}) = -64i$ ,  $A_5(\{2, 1, 3\}) = -784i$ ,  $A_5(\{5, 4, 1\}) = -11776i$ ;  $A_6(\{1, \frac{3}{2}, 2, \frac{5}{2}\}) = -\frac{968}{7}i$ ,  $A_6(\{4, 3, 2, 1\}) = -\frac{677376}{5}i$ ,  $A_6(\{\frac{5}{2}, 2, 3, \frac{7}{2}\}) = -\frac{1303400}{1}i$ ;  $A_7(\{1, \frac{3}{2}, 2, \frac{5}{2}, 3\}) = -\frac{1928}{5}i$ ,  $A_7(\{4, 3, \frac{5}{2}, 2, 1\}) = -\frac{426108561}{50}i$ , and the deliberately non-generic  $A_7(\{\frac{5}{2}, \frac{7}{2}, \frac{3}{2}, \frac{9}{4}, 3\}) = -\frac{12048407135}{8192}i$ . (I had also queued an  $n = 8$  batch; it bogged down and is not needed for the benchmark, so I trimmed it.) For  $n = 4$  the raw recursion is `Indeterminate` on every two-minus configuration, because the sector sits on exact zero-momentum internal channels — the removable  $0/0$  I had earlier mistaken for a hard obstruction. The same formula has  $m = n - 3 = 1$  there, so it reduces to

$$A_4 = 8i \omega_1 \omega_2 ([\omega_2^2]_+ - [\omega_2^2 - \omega_3^2]_+) = 8i \omega_1 \omega_2 \min(\omega_2^2, \omega_3^2),$$

which I report as the finite continuation, e.g. `freeW = {2, 3}` gives  $A_4 = -192i$  and  $\{\frac{5}{2}, \frac{7}{2}\}$  gives  $-\frac{875}{2}i$ .

**What I want to be honest about.** The answer is correct and exactly verified on the chart `MakeKinematics` actually produces — which is the kinematics the prompt specifies — so within the problem as posed this is

the closed form. But the route here was not clean. The prompt’s hint was actively misleading: it asserted a global rational function with simple poles and *no* piecewise/absolute-value structure, and it told me to avoid the very hierarchical regimes (one frequency crossing the others) where the chamber walls live. On my first pass I let that hint define what a valid answer could be, saw the true chamber structure at  $n = 4$ , and threw the benchmark out as inconsistent — the worst possible outcome, a correct observation turned into a wrong rejection. Only by refusing the hint’s shape on the second pass, resolving the Abs’s chamber by chamber, and reading the “poles” as spline corners did I recover the truncated-power formula. The hint’s promise of simple poles was false; the right object is a B-spline forced by  $|k_S|$ , exactly the thing the hint said could not occur.